

Xiangling Xu (许湘灵)

My Website

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ID Orcid



Research Interests: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

Education

- 2023 – now **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*
- 2020 – 2023 **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00
- 2016 – 2020 **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Gravity with Noncommutative Geometry, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

Publications and Preprints

- 1 F. M. Kalarde, S. Boreiri, **X. Xu**, L. Tendick, S. Beigi, P. Perinotti, T. Guaita, and M.-O. Renou, *Fermions are fundamentally more nonlocal than bosons*, Jun. 11, 2026. arXiv: 2606.12363 [quant-ph]. [URL: https://arxiv.org/abs/2606.12363](https://arxiv.org/abs/2606.12363).
- 2 **X. Xu**, M. Schötz, J. Wang, V. Magron, I. Klep, O. Fawzi, and M.-O. Renou, *The bulk spectral gap is semi-decidable: A convergent family of certified upper bounds*, Jun. 3, 2026. arXiv: 2606.03836 [quant-ph]. [URL: https://arxiv.org/abs/2606.03836](https://arxiv.org/abs/2606.03836).
- 3 F. M. Kalarde, **X. Xu**, and M.-O. Renou, *Comment on "quantum theory based on real numbers cannot be experimentally falsified": On the compatibility of physical principles with information theory for fermions*, Apr. 2026. arXiv: 2604.07425 [quant-ph]. [URL: https://arxiv.org/abs/2604.07425](https://arxiv.org/abs/2604.07425).
- 4 M.-O. Renou, **X. Xu**, and L. T. Ligthart, "Two convergent npa-like hierarchies for the quantum bilocal scenario," *Journal of Mathematical Physics*, vol. 67, no. 1, p. 012 203, Jan. 2026, ISSN: 0022-2488. [DOI: 10.1063/5.0211008](https://pubs.aip.org/aip/jmp/article-pdf/doi/10.1063/5.0211008/20865530/012203_1_5.0211008.pdf). eprint: https://pubs.aip.org/aip/jmp/article-pdf/doi/10.1063/5.0211008/20865530/012203_1_5.0211008.pdf.
- 5 M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, 2025. arXiv: 2509.25145 [quant-ph]. [URL: https://arxiv.org/abs/2509.25145](https://arxiv.org/abs/2509.25145).
- 6 I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, 2025. arXiv: 2507.17006 [quant-ph]. [URL: https://arxiv.org/abs/2507.17006](https://arxiv.org/abs/2507.17006).
- 7 **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph]. [URL: https://arxiv.org/abs/2505.22309](https://arxiv.org/abs/2505.22309).

- 8 X. Luo, X. Xu, and X. Wang, “On the kinematic morphology around haloes,” *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023. [DOI: https://doi.org/10.1093/mnras/stac3500](https://doi.org/10.1093/mnras/stac3500).
- 9 X. Xu, “Quantum nonlocality in bilocal networks: An operator algebraic perspective,” M.S. thesis, ETH Zurich, 2023. [DOI: https://doi.org/10.3929/ethz-b-000613670](https://doi.org/10.3929/ethz-b-000613670).

Talks & Conference Activities

Invited Talks

- Jun 2026  2026 SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom
- Dec 2025  Quantum Meets, online at <https://quantum-meets.github.io>
- Jul 2025  IWOTA 2025 Twente, Twente, Netherland
- Oct 2023  Saclay Quantum Seminar, Inria Saclay, France
- Jun 2025  IQC-PCQT-Quantum Saclay Workshop, Paris, France

Contributed Talks

- Aug 2026  QCrypt 2026, Ottawa, Canada
- Jan 2026  QIP 2026 Riga – 2 contributed talks, Riga, Latvia
- Nov 2025  3rd Colloquium GDR TeQ, Grenoble, France
- Oct 2025  YQIS25, Barcelona, Spain
- Sep 2024  Causalworlds 2024, Waterloo, Canada
- Nov 2023  Workshop Defi EQUIP, Lyon, France

Talks during group visits

- Sep 2026  Quiriosity, Inria & Télécom Paris & IP Paris
- Aug 2026  Ranyiliu Chen, Quantum Science Center of Huangdong-HongKong-Macao Greater Bay Area
- Jul 2026  Penghui Yao, Nanjing University
- Feb 2026  Jean-Daniel Bancal, CEA, Institut de physique théorique
- Dec 2025  Jie Wang, AMSS-CAS, Beijing
- Nov 2025  Thomas Vidick, EPFL
- Oct 2025  Jens Eisert, Freie Universität Berlin
- Feb 2025  QAT Team, ENS Paris
- Jul 2024  William Slofstra, University of Waterloo
- Dec 2023  David Gross, University of Cologne

Poster Sessions

- Aug 2026  CQCC & GQSF 2026, Shenzhen, China
- Nov 2024  2nd Colloquium GDR TeQ, Paris, France
-  YQIS24, Paris, France

Other Participation

- Mar 2025  828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory, Bad Honnef, Germany
- Sep 2024  VQF Vienna Quantum Foundations Conference, Vienna, Austria

Talks & Conference Activities (continued)

Nov 2023  *POP23 Workshop on Future Trends in Polynomial OPTimization*, Toulouse, France

Collaborations

Invited Group Visits

Oct-Nov 2026	 <i>David Gross</i> , University of Cologne Invited three-week visit
Jul 2026	 <i>Penghui Yao</i> , Nanjing University Invited one-week visit
Dec 2025 & May 2026	 <i>Jie Wang</i> , AMSS-CAS, Beijing Two invited one-week visits
Oct-Nov 2025	 <i>Thomas Vidick</i> , EPFL Invited two-week visit
Sep 2024	 <i>Igor Klep</i> , University of Ljubljana Invited two-week visit funded by Erasmus+ Mobility program
Jul 2024	 <i>William Slofstra</i> , Institute for Quantum Computing, University of Waterloo Invited two-week visit
Dec 2023	 <i>David Gross</i> , University of Cologne Invited one-week visit

Organizations

Mar 2025 - Now	 Coorganizer for <i>Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory</i> Coordinating the Gaspard Monge Visiting Professor program for Igor Klep Co-planning the seminar and the schedule for invited speakers
Mar 2025	 Coorganizer for <i>828. WE-Heraeus-Seminar at Bad Honnef</i> Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts Coordinating with the organizers and inviting some of the speakers
Feb 2024 - Now	 Coorganizer of <i>Project COMPUTE for Quanterra</i> Coordinating with the members Making the project poster and website
Sep 2023	 Coorganizer for <i>Saclay Quantum Seminar on NPA hierarchy</i> Planning and organizing the one-day program to bring all quantum groups at Saclay together

Awards and Scholarships

2026	 <i>Erasmus+ Mobility Program</i> , for visiting the University of Cologne
2024	 <i>Erasmus+ Mobility Program</i> , for visiting the University of Ljubljana
2020	 <i>McNab Undergraduate Scholarship</i> , University of Toronto
2017 - 2020	 <i>Dean's List Scholar</i> , University of Toronto
2018	 <i>The Class of 3To and Associates Scholarship in Mathematics and Physics</i> , University of Toronto
	 <i>International Exchange Award</i> , University of Toronto
	 <i>Bronze Medal</i> , 2017 University Physics Competition

Awards and Scholarships (continued)

- 2017  *Eileen Doman Award*, University of Toronto
-  *Summer Undergraduate Research Program*, Canadian Institute for Theoretical Astrophysics
- 2016  *President's Scholar of Excellence Program*, University of Toronto

Teaching Experience

Lecture

- Dec 2025  *Recent trends in parallel, distributed, and quantum computing*
QDCS Master, Université Paris-Saclay
- Nov 2024  *Recent trends in parallel, distributed, and quantum computing*
QDCS Master, Université Paris-Saclay

Teaching Assistance

- 2022-2023  *Linear Algebra I & II*
Department of Mathematics, ETH Zürich
- 2021-2022  *Analysis I & II*
Department of Mathematics, ETH Zürich

Skills

- Languages  Chinese (native), English (fluent), German (beginner)
- Coding  Python (Ncpolzsdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, L^AT_EX

Referees

Available on Request